

RESEARCH INTERESTS

Human-AI Interaction, Social Computing, Conversational Agents, Communication

SKILLS

Programming Languages: Python, R (proficient); C/C++, Java, SAS, SQL (intermediate)

Web Prototyping: JavaScript, HTML, CSS; backend frameworks (Django, Flask, Node.js)

Research Methods: Empirical studies, hypothesis testing, statistical modeling, inferential statistics, survey design, user studies, qualitative interviews, thematic analysis, web-based intervention development and prototyping; NLP and machine learning (familiar)

EDUCATION

University of Washington

Seattle, WA

Ph.D., Human Centered Design & Engineering

Sep 2023 - Jun 2028 (*expected*)

Advisor: Gary Hsieh & Sean A. Munson

University of Washington

Seattle, WA

M.S., Human Centered Design & Engineering (en route)

2023 - 2025

Ewha Womans University

Seoul, Korea

B.S., Computer Science & Engineering, Minor in Statistics

2019 - 2023

University of Toronto

Toronto, Canada

Undergraduate Exchange Program

2021 - 2021

PUBLICATIONS

Conference Papers

* indicates equal contribution

- [c.4] Meziah Ruby Cristobal*, **Hyeonjeong Byeon***, Alex Chen*, Ruoxi Shang*, Donghoon Shin*, Ruican Zhong*, Tony Zhou*, and Gary Hsieh. 2026. PaperTok: Exploring the Use of AI for Creating Short-form Videos for Research Communication. (*To appear*) In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '26)*. [paper]
- [c.3] **Hyeonjeong Byeon**, Uran Oh, and Gary Hsieh. 2026. Understanding the Effects of Conversational Agent Personality on the Credibility of LLM-Based Conversational Search. (*To appear*) In *Proceedings of the 2026 ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR '26)*. [paper]
- [c.2] Suhyeon Yoo, Georgianna Lin, **Hyeonjeong Byeon**, Amy Hwang, and Khai Truong. 2023. Understanding tensions in music accessibility through song signing for and with d/Deaf and Non-d/Deaf persons. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '23)*. [paper]
- [c.1] **Hyeonjeong Byeon***, Chaerin Lee*, Jeemin Lee, and Uran Oh. 2022. "A Voice that Suits the Situation": Understanding the Needs and Challenges for Supporting End-User Voice Customization. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '22)*. [paper]

Posters and Extended Abstracts

- [p.4] **Hyeonjeong Byeon**, Seungjin Ha, and Uran Oh. 2023. AVOCUS: A Voice Customization System for Online Personas. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. [paper]
- [p.3] **Hyeonjeong Byeon**. 2021. Voice Creator: Giving Customized Voice to the Voiceless for Online Communication. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility Student Research Competition (ASSETS' SRC 21)*. [paper]
- [p.2] **Hyeonjeong Byeon**, and Uran Oh. 2021. Comparing Performance of Pre-trained Bidirectional Language Models. In *Proceedings of HCI Korea Conference 2021*. [paper]
- [p.1] **Hyeonjeong Byeon**, Soohyun Yeo, and Uran Oh. 2021. Voice Creator: A Vocal Customization Web Application Prototype. In *2021 Korea Information Processing Society Spring Conference No. 28, Vol. 1*. [paper]

RESEARCH EXPERIENCES

Population Health Initiative, University of Washington Social Entrepreneurship Fellowship (PI: Weichao Yuwen) • Led the Virtual Study Assistant for potential research participants project	June 2025 - Aug 2025
Prosocial Computing Group, University of Washington Graduate Research Assistant (PI: Gary Hsieh) • Chatbot & Communication skills: Working on automated journalism and credibility frameworks to support scientific communication. • AI Journalism & Credibility: Working on automated journalism and credibility frameworks to support scientific communication. • Chatbot Personas & Credibility: Examining the effects of conversational agent design on search results.	May 2023 - Present
Intentional Traces Collective Lab, University of Washington Graduate Research Assistant (PI: Sean A. Munson) • Client–Clinician Match: Explored opportunities to support clinical communication and decision-making. • AI & Bibliotherapy: Investigated identity-matching interventions, focusing on the role of personal narratives and storytelling.	Jan 2024 - Present
Dynamic Graphics Project Lab, University of Toronto Undergraduate Research Assistant (PI: Khai N. Truong) • Creativity Support & Music Accessibility: Investigated opportunities for accessible music experiences for d/Deaf individuals using song signing.	Sep 2021 - Sep 2022
Human-Computer Interaction Lab, Ewha Womans University Undergraduate Research Assistant (PI: Uran Oh) • Voice Personas: Developed a search engine for labeled voice persona directories and a voice customization system. • Language Models & Patient Data: Applied bidirectional language models to classify electronic patient text data.	Jul 2020 - Sep 2023

GRANTS & AWARDS

Doctoral Student Research Grant University of Washington, Received grants to support projects for LLM-infused System Intervention Design	Dec 2025
Social Entrepreneurship Fellow Population Health Initiative, Leading the Virtual Study Assistant for Potential Research Participants Project	April 2025
Principal Candidate, Fulbright Graduate Student Program Award Korean-American Educational Commission, awarded up to \$80,000 for graduate studies (<i>declined</i>)	Sep 2023
Finalist, Doctoral Study Abroad Scholarship Korean Foundation for Advanced Studies (KFAS), selected as a finalist	Aug 2023
ITRC Student Creative Research Project South Korean Ministry of Science and ICT, awarded \$7,600 of research grant	March 2022 - Dec 2022
Capstone Poster Session Presentation Ewha Womans University, 2nd place in Computer Science department, co-awarded \$570	Dec 2022
SW Entrepreneurship Competition Ewha Womans University, 4th place, co-awarded \$75	Nov 2022
Google Student Travel Grants Google, awarded \$2,820 for attending ACM CHI 2022	May 2022
Ewha Future Design Scholarship Ewha Womans University, awarded \$3,240 for excellence in career path planning	Dec 2022, April 2022
3rd place (Bronze Medal), Student Research Competition ACM SIGACCESS Conference on Computers and Accessibility (ASSETS' 21) Student Research Competition, awarded \$200	Oct 2021
Presidential Science Scholarship Korea Student Aid Foundation (KOSAF), awarded cost of attendance for 2 years Additional \$2,200 per semester for academic excellence	June 2021
Best Paper Award Proceedings of HCI Korea Conference 2021, Seoul, South Korea	Jan 2021

Undergraduate Research Opportunity Program Research Grant

April 2022, June 2020

Ewha Womans University, awarded \$250, \$450 respectively

HOKMA Career Path Exploration Scholarship Grant

Oct 2019

Ewha Womans University, co-awarded \$1,100 for excellence in CS career path exploration activities

PATENT**Systems and Methods for Customizing Voices for Personas in Virtual Spaces**

Dec 2022

Uran Oh, Hyeonjeong Byeon, Seungjin Ha, 10-2022-0179981. *Patent pending.***AVOTAR: A Voice Customization System for Online Personas**

Nov 2022

Uran Oh, Hyeonjeong Byeon, Seungjin Ha, South Korea, P2022-0160KR. *Patent pending.*

TEACHING EXPERIENCE

Visual Communication (HCDE 508)

Designing For Behavior Change (HCDE 538)

Usability Studies (HCDE 517)

User Experience Prototyping Techniques (HCDE 451)

Introduction to User Research (HCDE 313)

SERVICE**Review Service**

2024 -

ACM CHI (2026 -), ACM CHI Late-Breaking Work (2024 -), IASDR (2025)

Student Admission Reviewer

Dec 2023

Participated as a student reviewer for UW HCDE Ph.D. admission

Student Volunteer, The Blue Tree Foundation (NGO)

Feb 2022 - Feb 2022

Researched and translated articles about school violence and cyberbullying for campaigns

Member, EDOC (Ewha DO Coding)

Jan 2020 - Jun 2020

Prepared for ACM-ICPC and participated in group study sessions focusing on data structures and algorithms

Student Volunteer, Seoul Metropolitan Office of Education

Sep 2019 - Nov 2019

Taught programming to North Korean defector students as part of the "Unplugged Coding" program

Member, Like Lion

Mar 2019 - Dec 2019

Completed a year-long full-stack development course.

Back-end development of the university festival web application [Web Demo] (*closed*)

RELEVANT SELECTIVE COURSEWORK**Natural Language Processing (CSE 517)**

Reproduced an EMNLP 2024 paper and conducted additional generalizability experiments [code]

Core Concepts in Decision Making (PSYCH 521)

Dual-process theories of reasoning, heuristics and biases, nudging, and decision-making with AI support

Applied Social Statistics (SOC 504)

Regression modeling, hypothesis testing, and applied statistical inference

Communication in Small Groups (COM 577)

Seminars on group interaction, social influence, norms, trust, identity, and performance